flows through and deposits its load on its either side. Example, Tiber River of Italy

4.7 Glacier

A glacier is a huge mass of ice that moves slowly along the mountain slope. The term "glacier" comes from the French word *glace* which means ice. Glaciers are often called "rivers of ice". It forms where the accumulation of snow exceeds its ablation over many years.

The places where the snow lies for the whole year are called **snowfields**. The imaginary line above which there is a permanent snowfield is known as **the snow line**. The snowfields are always situated above the snow line. The snow line differs according to latitude, amount of snowfall, the direction of the wind and the physical features of the region. Snow starts melting below the snow line. Under the pressure of the upper layers, the lower layers of the snow field begin to melt causing the mass of snow to move down slope as glacier. Glacier moves at an average speed of 1meter per day. Over 96 percent of the glaciers occur in Antarctica and Greenland.



The world's largest glacier is the Lambert Glacier in Antarctica, more than 96 km wide and 435 km long and 2,500 metres deep..

4.7.1 Types of Glaciers

The Glaciers are of three types. They are;

1. Continental Glaciers
2. Ice Caps
3. Mountain and Valley Glaciers

1. Continental Glaciers

The continental glaciers are found in polar regions. In these areas, all the precipitation is in the form of snow. The snow that falls from year to year gradually gets accumulated. As a result, these regions are covered by an extensive ice mass. This is known as ice sheet or continental glacier. It is estimated that the maximum thickness of the ice sheets of Greenland is 3,400 meter, while the maximum thickness of the ice sheet of Antarctica is 4,776 meter. Sometimes, the ends of the ice sheet projects outwards over the sea. The waves of the sea strike against them and break the ice sheets into blocks of floating ice known as Icebergs.

2. Ice Caps

It is the covering of snow and ice on the oceans of poles. The ice caps can cover vast areas with the extensive accumulation of snow and ice. Example, Svartissen ice cap in Northern Norway.

3. Mountain and Valley Glaciers

These are also known as Alpine glaciers. They flow like tongues of ice down through the mountain valleys from the ice caps. The piedmont glaciers form continuous ice sheets at the base of mountains. The valley glaciers or Alpine glaciers are found in higher regions of the Himalayas and on all such high mountain ranges of the world.

4.7.2 Characteristics of Glaciers

A moving ice mass or glacier possesses certain characteristics of movement, speed, and surface structure. The rate of movement of the glacier is very slow.

The rate of movement or the speed of a glacier depends upon the size of the glacier

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